

Functions

Question1

The domain of the real valued function $f(x) = \frac{3}{4-x^2} + \log_{10}(x^3 - x)$ is

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Options:

A.

$$(1, 2) \cup (2, \infty)$$

B.

$$(-1, 0) \cup (1, 2)$$

C.

$$(-1, 0) \cup (1, 2) \cup (2, \infty)$$

D.

$$(-\infty, -1) \cup (1, 2) \cup (2, \infty)$$

Answer: C

Solution:

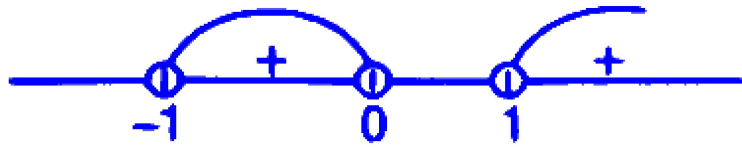
$$f(x) = \frac{3}{4-x^2} + \log_{10}(x^3 - x)$$

$$\text{for } \frac{3}{4-x^2}, 4-x^2 \neq 0$$

$$\Rightarrow x \neq \pm 2$$

$$\text{and for } \log_{10}(x^3 - x),$$

$$x^3 - x > 0 \Rightarrow x(x^2 - 1) > 0$$



and $x \neq 2$

Here, $x \in (-1, 0) \cup (1, 2) \cup (2, \infty)$

Question2

A real valued function $f : A \rightarrow B$ defined by $f(x) = \frac{4-x^2}{4+x^2} \forall x \in A$ is a bijection. If $-4 \in A$, then $A \cap B =$

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Options:

A.

$(-1, 1]$

B.

$[0, 1]$

C.

$[0, \infty)$

D.

$(-1, 0]$

Answer: D

Solution:

$\because f(x) = \frac{4-x^2}{4+x^2}, \forall x \in A$ and $f(x)$ is bijective So, Domain of $f(x) 4 + x^2 \neq 0$ which is always true. therefore, Domain of $f(x) \in R$ Lets check if $f(x)$ be injective

$$\begin{aligned}
f(x_1) &= f(x_2) \\
\Rightarrow \frac{4 - x_1^2}{4 + x_1^2} &= \frac{4 - x_2^2}{4 + x_2^2} \\
\Rightarrow 16 + 4x_2^2 - 4x_1^2 - x_1^2 x_2^2 \\
&= 16 + 4x_1^2 - 4x_2^2 - x_1^2 x_2^2 \\
\Rightarrow x_1^2 &= x_2^2 \Rightarrow x_2 = \pm x_1
\end{aligned}$$

for $f(x)$ to be injective.

We must have $x_1 = x_2$ and $-4 \in A$ therefore, domain must be $(-\infty, 0]$ Now, determine the range

$$\text{Let, } y = \frac{4-x^2}{4+x^2}$$

$$\Rightarrow y(4 + x^2) = 4 - x^2 \Rightarrow 4y + yx^2 = 4 - x^2$$

$$\Rightarrow x^2 = \frac{4(1-y)}{1+y}$$

$$\because x^2 \geq 0$$

$$\text{So, } \frac{4(1-y)}{1+y} \geq 0$$

$$\text{So, } 1 - y \geq 0 \text{ and } 1 + y > 0$$

$$\text{therefore, range of } f(x) \in (-1, 1]$$

$$\text{that means, } A \subseteq (-\infty, 0]$$

$$\text{and } B = (-1, 1]$$

$$\text{Hence, } A \cap B = (-1, 0]$$

Question3

Let $f(x) = x^2 + 2bx + 2c^2$ and $g(x) = -x^2 - 2cx + b^2$. $x \in R$. If b and c are non-zero real numbers such that $\min f(x) > \max g(x)$, then $\left| \frac{c}{b} \right|$ lies in the interval

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Options:

A.

$$\left(\frac{1}{2}, \frac{1}{\sqrt{2}} \right)$$

B.

$$\left(\frac{1}{\sqrt{2}}, \sqrt{2}\right)$$

C.

$$(\sqrt{2}, \infty)$$

D.

$$(0, 1)$$

Answer: C

Solution:

$f(x) = x^2 + 2bx + 2c^2$ is a quadratic, opening upward

So, minimum at $(x) = \frac{-2b}{2} = -b$

then, $f(-b) = b^2 - 2b^2 + 2c^2 - b^2 + 2c^2$
and $g(x) = -x^2 - 2cx + b^2$ is quadratic opening downward.

So, maximum of $g(x)$ at $x = \frac{-(-2c)}{2(-1)} = -c$

then, $g(-c) = -c^2 + 2c^2 + b^2 = c^2 + b^2$

and given that

$$\min f(x) > \max g(x)$$

$$\Rightarrow -b^2 + 2c^2 > c^2 + b^2$$

$$\Rightarrow c^2 > 2b^2 \Rightarrow \left(\frac{c}{b}\right)^2 > 2$$

$$\Rightarrow \left|\frac{c}{b}\right| > \sqrt{2}$$

Hence, $\left|\frac{c}{b}\right| \in (\sqrt{2}, \infty)$

Question4

Let $[x]$ represent the greatest integer less than or equal to

x , $\{x\} = x - [x] \sqrt{2} = 1.414$ and $\sqrt{3} = 1.732$. If

$f(x) = \left\{x + \left[\frac{x}{1+x^2}\right]\right\}$ is a real valued function, then

$$f(\sqrt{2}) + f(-\sqrt{3}) =$$

Options:

A.

0.682

B.

0.318

C.

0.146

D.

1.146

Answer: A

Solution:

We have,

$$f(x) = \left\{ x + \left[\frac{x}{1+x^2} \right] \right\}$$

$$f(\sqrt{2}) = \left\{ \sqrt{2} + \left[\frac{\sqrt{2}}{5} \right] \right\} = \{\sqrt{2}\} = \sqrt{2} - 1$$

$$\text{and } f(-\sqrt{3}) = \left\{ -\sqrt{3} + \left[\frac{-\sqrt{3}}{4} \right] \right\}$$

$$= \{-\sqrt{3} - 1\}$$

$$\Rightarrow f(-\sqrt{3}) = 2 - \sqrt{3}$$

$$\therefore f(\sqrt{2}) + f(-\sqrt{3}) = \sqrt{2} - 1 + 2 - \sqrt{3}$$

$$= 1 + \sqrt{2} - \sqrt{3} = 1 + 1.414 - 1.732$$

$$= 2.414 - 1.732 = 0.682$$

Question5

If the range of the function $f(x) = -3x - 3$ is $\{3, -6, -9, -18\}$, then which one of the following is not in the domain of f ?

Options:

A.

-1

B.

-2

C.

2

D.

5

Answer: A

Solution:

The function given is $f(x) = -3x - 3$.

The range of the function is given as $\{3, -6, -9, -18\}$.

To find the elements in the domain corresponding to this range, we need to find the x values for which $f(x)$ equals each value in the range.

We set $f(x)$ equal to each value in the range and solve for x :

1. Set $f(x) = 3$:

$$-3x - 3 = 3$$

$$-3x = 3 + 3$$

$$-3x = 6$$

$$x = \frac{6}{-3}$$

$$x = -2$$

2. Set $f(x) = -6$:

$$-3x - 3 = -6$$

$$-3x = -6 + 3$$

$$-3x = -3$$

$$x = \frac{-3}{-3}$$

$$x = 1$$

3. Set $f(x) = -9$:

$$-3x - 3 = -9$$

$$-3x = -9 + 3$$

$$-3x = -6$$

$$x = \frac{-6}{-3}$$

$$x = 2$$

4. Set $f(x) = -18$:

$$-3x - 3 = -18$$

$$-3x = -18 + 3$$

$$-3x = -15$$

$$x = \frac{-15}{-3}$$

$$x = 5$$

So, the domain of f corresponding to the given range $\{3, -6, -9, -18\}$ is the set of x values $\{-2, 1, 2, 5\}$.

Now we need to check the given options to see which one is not in this domain:

A) -1: Is -1 in $\{-2, 1, 2, 5\}$? No.

B) -2: Is -2 in $\{-2, 1, 2, 5\}$? Yes. ($f(-2) = 3$)

C) 2: Is 2 in $\{-2, 1, 2, 5\}$? Yes. ($f(2) = -9$)

D) 5: Is 5 in $\{-2, 1, 2, 5\}$? Yes. ($f(5) = -18$)

The value -1 is not in the calculated domain.

The final answer is $\boxed{-1}$.

Question6

$[t]$ denotes the greatest integer function and $[t - m] = [t] - m$ when $m \in \mathbb{Z}$. If $k = 2[2x - 1] - 1$ and $3[2x - 2] + 1 = 2[2x - 1] - 1$, then the range of $f(x) = [k + 5x]$ is

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Options:

A.

$\{7, 8, 9\}$

B.

$\{4, 5, 6\}$

C.

$\{5, 6, 7\}$

D.

$\{6, 7, 8\}$

Answer: D

Solution:

$$\begin{aligned}\therefore [2x - 2] &= [(2x - 1) - 1] \\ &= [2x - 1] - 1\end{aligned}$$

$$\text{let, } [2x - 1] = a$$

$$\Rightarrow [2x - 2] = a - 1$$

$$\text{so, } 3(a - 1) + 1 = 2a - 1$$

$$\Rightarrow 3a - 3 + 1 = 2a - 1 \Rightarrow a = 1$$

$$\text{thus } [2x - 1] = 1$$

$$\Rightarrow 1 \leq (2x - 1) < 2 \Rightarrow 2 \leq 2x < 3$$

$$\Rightarrow 1 \leq x < 1.5$$

$$\text{so, } x \in [1, 1.5)$$

$$\text{Now, } k = 2[2x - 1] - 1 = 2(1) - 1 = 1$$

$$\text{for } x \in [1, 1.5)$$

$$f(x) = [k + 5x] = [1 + 5x]$$

$$\text{at } x = 1, f(x) = [6] = 6 \quad (\text{minimum})$$

$$\begin{aligned}\text{at } x = 1.5, f(x) &= [1 + 5 \times 1.5] \\ &= [8.5] = 8 \quad (\text{maximum})\end{aligned}$$

so, $f(x)$ can be 6, 7, 8

Hence, the range of $f(x) = \{6, 7, 8\}$

Question7

If $f(x) = (x + 1)^2 - 1$, $x \geq -1$, then $\{x \mid f(x) = f^{-1}(x)\}$ is

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Options:

A.

$$\{0, -1\}$$

B.

$$\{-1, 0, 1\}$$

C.

$$\left\{-1, 0, \frac{-3+\sqrt{3}i}{2}, \frac{-3-\sqrt{3}i}{2}\right\}$$

D.

an empty set

Answer: A

Solution:

$$\because y = f(x) = (x+1)^2 - 1$$

$$\Rightarrow y + 1 = (x+1)^2$$

$$\Rightarrow x = \sqrt{y+1} - 1$$

thus, $f^{-1}(x) = \sqrt{x+1} - 1$, defined for

$$x \geq -1$$

Now, $f(x) = f^{-1}(x)$

$$\Rightarrow (x+1)^2 = \sqrt{x+1}$$

$$\text{let } x+1 = t$$

$$\Rightarrow t^4 = t \Rightarrow t(t^3 - 1) = 0$$

therefore, $t = 0$

$$\Rightarrow x+1 = 0 \Rightarrow x = -1$$

$$\text{or } t = 1$$

$$\Rightarrow x+1 = 1 \Rightarrow x = 0$$

$$\text{Hence, } \{x \mid f(x) = f^{-1}(x)\} \\ = \{0, -1\}$$

Question8

Consider the following statements.

Statement I	A function $f : A \rightarrow B$ is said to be one-one if and only if $f(x) \neq f(y) \Rightarrow x \neq y$
Statement II	A relation $f : A \rightarrow B$ is said to be a function if $x \neq y \Rightarrow f(x) \neq f(y)$

Then, which one of the following is true?

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Options:

A.

Only statement I is true.

B.

Only statement II is true.

C.

Both Statement I and Statement II are true.

D.

Neither Statement I nor Statement II is true.

Answer: A

Solution:

It is obvious. Since, f is said to be one-one if $f(x) = f(y) \Rightarrow x = y$ or $f(x) \neq f(y) \Rightarrow x \neq y$. Also, in a function, one image may have more than one pre-image.

Question9

The set of all real values of x for which $f(x) = \sqrt{\frac{|x|-2}{|x|-3}}$ is a well defined function is

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Options:

A.

$$(-3, -2] \cup (2, 3]$$

B.

$$R - [-3, -2) \cup (2, 3]$$

C.

$$R - [-3, 3]$$

D.

$$(-3, 3)$$

Answer: B

Solution:

$$f(x) = \sqrt{\frac{|x|-2}{|x|-3}}$$

$$\Rightarrow \frac{|x|-2}{|x|-3} \geq 0$$

$$\text{Case I } |x| - 2 \geq 0 \text{ and } |x| - 3 > 0$$

$$\Rightarrow |x| \geq 2 \text{ and } |x| > 3$$

$$\Rightarrow |x| > 3$$

$$\Rightarrow x > 3 \text{ or } x < -3$$

$$\text{Case II } |x| - 2 \leq 0 \text{ and } |x| - 3 < 0$$

$$|x| \leq 2 \text{ and } |x| < 3$$

$$\Rightarrow |x| \leq 2$$

$$-2 \leq x \leq 2$$

$$|x| - 3 \neq 0 \Rightarrow |x| \neq 3$$

$$\Rightarrow x \neq 3 \text{ and } x \neq -3$$

Combining these, we get

$$(-\infty, -3) \cup [-2, 2] \cup (3, \infty)$$

Finally we write

$$R - [-3, -2) \cup (2, 3]$$

Question10

Let $f : N \rightarrow N$ be a function such that $f(x + y) = f(x) + f(y) + xy$ for every $x, y \in N$. If $f(1) = 2$, then $\sum_{k=0}^{10} f(k) =$

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Options:

A.

1650

B.

275

C.

550

D.

1025

Answer: B

Solution:

$$f(x + y) = f(x) + f(y) + xy \quad \dots (i)$$

$$x, y \in N$$

From Eq. (i),

$$f(1) = 2 = \frac{1(1 + 3)}{2}$$

$$x = 1, y = 1$$

$$f(2) = f(1) + f(1) + 1$$

$$f(2) = 5 = \frac{2(2 + 3)}{2}$$

From Eq. (i), $x = 1, y = 2$

$$f(3) = f(1) + f(2) + 2$$

$$= 2 + 5 + 2 = 9 = \frac{3(3+3)}{2}$$

$$f(4) = f(2) + f(2) + 4$$

$$= 14 = \frac{4(4+3)}{2}$$

$$\Rightarrow f(x) = \frac{n(n+3)}{2} = \frac{n^2}{2} + \frac{3n}{2}$$

$$\sum_{k=0}^{10} f(k) = \sum_{k=1}^{10} \frac{k(k+3)}{2} = \frac{1}{2} \sum_{k=1}^{10} k^2 + \frac{3}{2} \sum_{k=1}^{10} k$$

$$= \frac{1}{2}(385 + 165) = 275$$

Question11

If a real valued function $f : [-1, 2] \rightarrow B$ defined by

$$f(x) = \begin{cases} 1-x, & \text{when } -1 \leq x \leq 1 \\ x-1, & \text{when } 1 < x \leq 2 \end{cases}$$

is a surjection, then $B =$

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Options:

A.

$$[-1, 2]$$

B.

$$[-1, 1]$$

C.

$$[0, 2]$$

D.

$$[0, 1]$$

Answer: C

Solution:

$$f(x) = \begin{cases} 1-x, & x \in [-1, 1] \\ x-1, & x \in (1, 2] \end{cases}$$

$$-1 \leq x \leq 1$$

$$+1 \geq -x \geq -1$$

$$+2 \geq 1-x \geq +0$$

$$x \in [-1, 1]$$

$$f(x) \in [0, 2] \quad \dots (i)$$

$$\text{Now } x \in (1, 2], f(x) \in (0, 1] \quad \dots (ii)$$

From Eqs. (i) and (ii) range $\equiv [0, 2]$

Question12

The sum of the least positive integer and the greatest negative integer in the range of the function $f(x) = \frac{x^2-5x+7}{x^2-5x-7}$ is

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Options:

A.

0

B.

1

C.

2

D.

-1

Answer: B

Solution:

$$\text{Given, } f(x) = \frac{x^2-5x+7}{x^2-5x-7} = y$$

$$\Rightarrow (y-1)x^2 - 5(y-1)x - (7y+7) = 0$$

for $x \in R$

$$D \geq 0$$

$$\Rightarrow 25(y-1)^2 + 4(y-1)(7y+7) \geq 0$$

$$\Rightarrow (y-1)[25y - 25 + 28y + 28] \geq 0$$

$$(y-1)(53y+3) \geq 0$$

$$y \in \left(-\infty, \frac{-3}{53}\right] \cup (1, \infty) \text{ at } y=1, x \text{ is not}$$

possible

$$\therefore \text{Required sum} = 2 + (-1) = 1$$

Question13

The interval in which the curve represented by $f(x) = 2x + \log\left(\frac{x}{2+x}\right)$ is

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Options:

A.

$$(-\infty, 0)$$

B.

$$(-2, \infty)$$

C.

$$(-\infty, -2) \cup (0, \infty)$$

D.

$$(-2, 0)$$

Answer: C

Solution:

$$f(x) = 2x + \log\left(\frac{x}{2+x}\right), \frac{x}{2+x} > 0 \text{ and } \frac{x}{2+x} \neq 0 \Rightarrow x \neq 0$$

$$x \in (-\infty, -2) \cup (0, \infty)$$

$$f'(x) = 2 + \left(\frac{2+x}{x} \right) \cdot \left(\frac{2}{(2+x)^2} \right)$$

$$= 2 + \frac{2}{x(x+2)}, x \neq -2$$

$$f'(x) = 0, 2 = \frac{-2}{x(x+2)}$$

$$x^2 + 2x + 1 = 0$$

$$(x+1)^2 = 0 \Rightarrow x = -1$$

$$x = -1 \notin (-\infty, -2) \cup (0, \infty)$$

$$f'(x) > 0 \text{ for } x \in (-\infty, -2) \cup (0, \infty)$$

So, required interval,

$$(-\infty, -2) \cup (0, \infty)$$

Question14

The set of real values of x such that $f(x) = \sqrt{\frac{[x]-1}{[x]^2-[x]-6}}$ is a real valued function is

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Options:

A.

$$[1, \infty)$$

B.

$$(-\infty, -2) \cup [4, \infty)$$

C.

$$[-1, 3)$$

D.

$$[-1, 2) \cup [4, \infty)$$

Answer: D

Solution:

$$f(x) = \sqrt{\frac{[x] - 1}{[x]^2 - [x] - 6}}$$

$$f(x) \text{ is defined when } \frac{[x] - 1}{[x]^2 - [x] - 6} \geq 0$$

$$\frac{[x] - 1}{([x] - 3)([x] + 2)} \geq 0$$

$$\Rightarrow -2 < [x] \leq 1 \text{ or } [x] \geq 3$$

$$\Rightarrow -1 \leq x < 2 \text{ or } x \geq 4 \text{ or } x \in [-1, 2) \cup [4, \infty)$$

Question15

If a function $f : Z \rightarrow Z$ is defined by $f(x) = x - (-1)^x$, then $f(x)$ is

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Options:

A.

one-one, but not onto

B.

onto but not one-one

C.

both one-one and onto

D.

neither one-one nor onto

Answer: C

Solution:

$$f(x) = x - (-1)^x = \begin{cases} x - 1, & x \in \text{even} \\ x + 1, & x \in \text{odd} \end{cases}$$

$f(x)$ is a linear function. Hence, it is one-one and its range is integer for integral value of x so it is onto.

$\therefore$ It is both one-one and onto.

Question16

Domain of the real valued function $f(x) = \log(x^2 - 1) + x \coth^{-1} x$ is

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Options:

A.

R

B.

$(-1, 1)$

C.

$R - [-1, 1]$

D.

$R - [0, 1]$

Answer: C

Solution:

$f(x) = \log(x^2 - 1) + x \cot h^{-1} x \log(x^2 - 1)$ is defined when $x^2 - 1 > 0 \Rightarrow x^2 > 1$

$-\infty < x < -1$ or $1 < x < \infty$ And $\cot h^{-1} x$ is defined when $|x| > 1 \Rightarrow -\infty < x < -1$ or $1 < x < \infty$

$\therefore$ Domain of $f(x)$ is $x \in (-\infty, -1) \cup (1, \infty)$ or $R - [-1, 1]$

Question17

The domain and range of a real valued function $f(x) = \cos x - 3$ are respectively

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Options:

A.

$$\mathbb{R} \setminus \{0\} \text{ and } [-1, 1]$$

B.

$$\mathbb{R} \text{ and } [-1, 1]$$

C.

$$\mathbb{R} \setminus \{0\} \text{ and } [-4, -2]$$

D.

$$\mathbb{R} \text{ and } [-4, -2]$$

Answer: D

Solution:

$$\text{We have } f(x) = \cos x - 3$$

$$\text{Domain} = \mathbb{R}$$

$$\text{Range } -1 \leq \cos x \leq 1$$

$$-1 - 3 \leq \cos x - 3 \leq 1 - 3$$

$$-4 \leq f(x) \leq -2$$

$$\text{Range} = [-4, -2]$$

Question18

If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are two functions defined by $f(x) = 2x - 3$ and $g(x) = 5x^2 - 2$, then the least value of the function $(g \circ f)(x)$ is

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Options:

A.

B.

2

C.

-4

D.

4

Answer: A

Solution:

Given, $f(x) = 2x - 3$ and $g(x) = 5x^2 - 2$
 $g \circ f(x) = 5[f(x)]^2 - 2 = 5(2x - 3)^2 - 2$
 $\therefore$ Least value of $g \circ f(x) = -2$

Question19

If $A \subseteq \mathbb{Z}$ and the function $f : A \rightarrow \mathbb{R}$ is defined by
 $f(x) = \frac{1}{\sqrt{64 - (0.5)^{24+x-x^2}}}$, then the sum of all absolute value of elements
of A is

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Options:

A. 36

B. 5

C. 25

D. 11

Answer: C

Solution:

We have, $f(x) = \frac{1}{\sqrt{64 - (0.5)^{24+x-x^2}}}$

$f(x)$ will be defined, if $64 - (0.5)^{24+x-x^2} > 0$

$$= 64 > (0.5)^{24+x-x^2}$$

$$= 64 > 2^{x^2-x-24}$$

$$= 2^6 > 2^{x^2-x-24}$$

$$= x^2 - x - 24 < 6$$

$$= x^2 - x - 30 < 0$$

$$= (x+5)(x-6) < 0$$

$$= -5 < x < 6$$

$$\therefore x = -4, -3, -2, -1, 0, 1, 2, 3, 4, 5$$

Then, $A = \{-4, -3, -2, -1, 0, 1, 2, 3, 4, 5\}$

Sum of all absolute values of A is

$$= 4 + 3 + 2 + 1 + 0 + 1 + 2 + 3 + 4 + 5$$

$$= 25$$

Question20

Which of the following function are odd?

I. $f(x) = x \left(\frac{e^x - 1}{e^x + 1} \right)$

II. $f(x) = k^x + k^{-x} + \cos x$

III. $f(x) = \log \left(x + \sqrt{x^2 + 1} \right)$

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Options:

A. II

B. I II

C. III

D. I

Answer: C

Solution:

We have, $f(x) = \log(x + \sqrt{x^2 + 1})$

$$\begin{aligned}\Rightarrow f(-x) &= \log(-x + \sqrt{x^2 + 1}) \\&= \log\left[\sqrt{x^2 + 1} - x \times \frac{\sqrt{x^2 + 1} + x}{\sqrt{x^2 + 1} + x}\right] \\&= \log\left[\frac{x^2 + 1 - x^2}{\sqrt{x^2 + 1} + x}\right] \\&= \log\left(\frac{1}{\sqrt{x^2 + 1} + x}\right) \\&= -\log(x + \sqrt{x^2 + 1}) \\&= -f(x)\end{aligned}$$

$\therefore f(x)$ is an odd function.

Question21

The range of the real valued function $f(x) = \frac{15}{3 \sin x + 4 \cos x + 10}$ is

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Options:

- A. $[0, 3]$
- B. $[-1, 3]$
- C. $[1, 3]$
- D. $[-1, 1]$

Answer: C

Solution:

We need to determine the range of the function:

$$f(x) = \frac{15}{3 \sin x + 4 \cos x + 10}$$

Start with the expression inside the denominator:

$$3 \sin x + 4 \cos x$$

The range of this expression is given by:

$$-\sqrt{3^2 + 4^2} \leq 3 \sin x + 4 \cos x \leq \sqrt{3^2 + 4^2}$$

Calculating the bounds:

$$-\sqrt{9 + 16} \leq 3 \sin x + 4 \cos x \leq \sqrt{9 + 16}$$

$$-5 \leq 3 \sin x + 4 \cos x \leq 5$$

Adding 10 to the entire inequality, we have:

$$5 \leq 3 \sin x + 4 \cos x + 10 \leq 15$$

The function $f(x)$ can then be rewritten as:

$$f(x) = \frac{15}{3 \sin x + 4 \cos x + 10}$$

Therefore, the minimum value of the denominator is 5, and the maximum is 15.

The range of $f(x)$ can be calculated as follows:

Minimum value of $f(x)$:

$$f(x)_{\min} = \frac{15}{15} = 1$$

Maximum value of $f(x)$:

$$f(x)_{\max} = \frac{15}{5} = 3$$

Hence, the range of the function $f(x)$ is $[1, 3]$.

Question22

Define the function, f, g and h from R to R such that

$$f(x) = x^2 - 1, g(x) = \sqrt{x^2 + 1} \text{ and } h(x) = \begin{cases} 0, & \text{if } x \leq 0 \\ x, & \text{if } x \geq 0 \end{cases} \text{ consider}$$

the following statements

(i) fog is invertible

(ii) h is an identify function

(iii) $f \circ g$ is not invertible

(iv) $(h \circ f \circ g)x = x^2$

Then, which one of the following is true ?

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Options:

A. II, IV

B. II, III

C. III, IV

D. I, II

Answer: C

Solution:

Given the functions:

$$f(x) = x^2 - 1$$

$$g(x) = \sqrt{x^2 + 1}$$

$$h(x) = \begin{cases} 0, & \text{if } x \leq 0 \\ x, & \text{if } x \geq 0 \end{cases}$$

Let's analyze the statements:

Function Composition:

$$f \circ g = f(g(x)) = f(\sqrt{x^2 + 1}) = (\sqrt{x^2 + 1})^2 - 1 = x^2 + 1 - 1 = x^2.$$

Since $f \circ g = x^2$, it is clear that this function is not one-to-one because different values of x (such as x and $-x$) yield the same x^2 . Therefore, $f \circ g$ is not invertible.

Composition involving h :

$$h \circ f \circ g = h(f(g(x))) = h(x^2).$$

The function $h(x^2)$ has the rule:

$$h(x^2) = \begin{cases} 0, & \text{if } x \leq 0 \\ x^2, & \text{if } x \geq 0 \end{cases}$$

Since x^2 is always non-negative and $h(x^2) = x^2$ for $x \geq 0$, this simplifies to $h(x^2) = x^2$ across non-negative ranges, reinforcing that $h(x^2)$ is equivalent to x^2 .

Considering these observations, statements (III) and (IV) are true:

Statement (III): $f \circ g$ is not invertible.

Statement (IV): $h \circ f \circ g = x^2$.

Thus, statements (III) and (IV) are correct.

Question23

The domain of the real valued function $f(x) = \sqrt{9 - \sqrt{x^2 - 144}}$ is

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Options:

A. $[-15, -12] \cup [12, 15]$

B. $(-\infty, -12] \cup [12, \infty)$

C. $[-15, 15]$

D. $[-12, -12]$

Answer: A

Solution:

For real values of $f(x)$

$$\sqrt{9 - \sqrt{x^2 - 144}} \geq 0$$

On squaring both sides, we get

$$\begin{aligned} 9 - \sqrt{x^2 - 144} &\geq 0 \\ 9 &\geq \sqrt{x^2 - 144} \end{aligned}$$

Again, squaring on both sides, we get

$$\begin{aligned} 9^2 &\geq x^2 - 144 \\ 81 + 144 &\geq x^2 \\ 225 &\geq x^2 \\ x^2 &\leq 225 \end{aligned}$$

On taking principal square root

$$|x| \leq 15$$

$$\text{i.e. } x \leq 15 \text{ or } x \geq -15$$

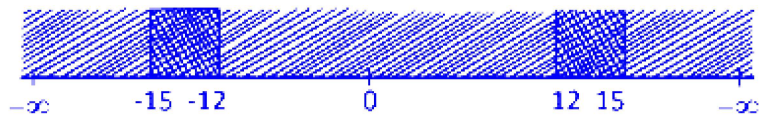
$$\begin{aligned} \text{Now, } x^2 - 144 &\geq 0 \\ x^2 &\geq 144 \end{aligned}$$

On taking principal square root

$$\sqrt{x^2} \geq \sqrt{12^2}$$

$$|x| \geq 12$$

i.e. $x \leq -12$ or $x \geq 12$



We can see on number line the domain of the given function will be cross shadow area.

i.e. $[-15, -12] \cup [12, 15]$

Question24

The real valued function $f : R \rightarrow \left[\frac{5}{2}, \infty\right)$ defined by $f(x) = |2x + 1| + |x - 2|$ is

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Options:

- A. One - one function but not onto
- B. Onto function but not one - one
- C. Bijection
- D. Neither one - one function not onto

Answer: B

Solution:

Given function $f : R \rightarrow \left[\frac{5}{2}, \infty\right]$ is defined by

$$f(x) = |2x + 1| + |x - 2|$$

Consider the graph

There are two points, where the expressions inside the absolute values change signs.

$$x = -\frac{1}{2} \text{ or } -0.5 \text{ and } x = 2$$

If $x < -0.5$

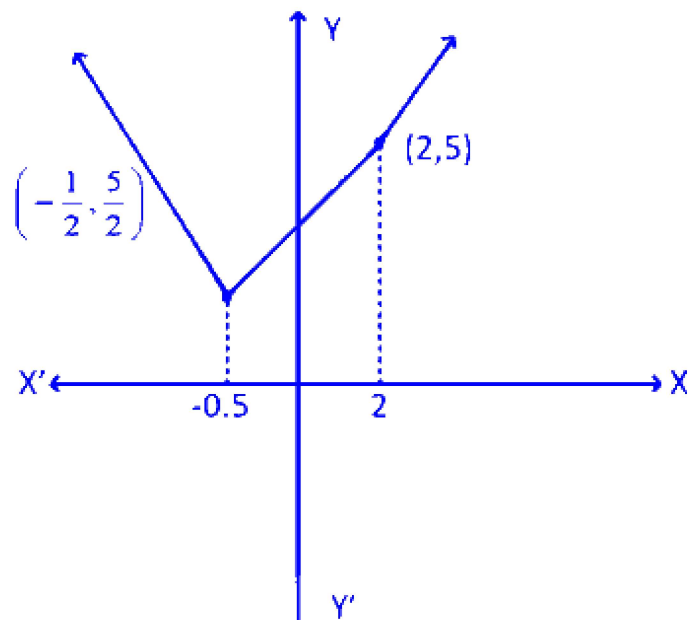
$$\begin{aligned} f(x) &= -2x - 1 - x + 2 \\ &= -3x + 1 \end{aligned}$$

If $-0.5 < x < 2$

$$f(x) = 2x + 1 - x + 2 = x + 3$$

If $x > 2$

$$f(x) = 2x + 1 + x - 2 = 3x - 1$$

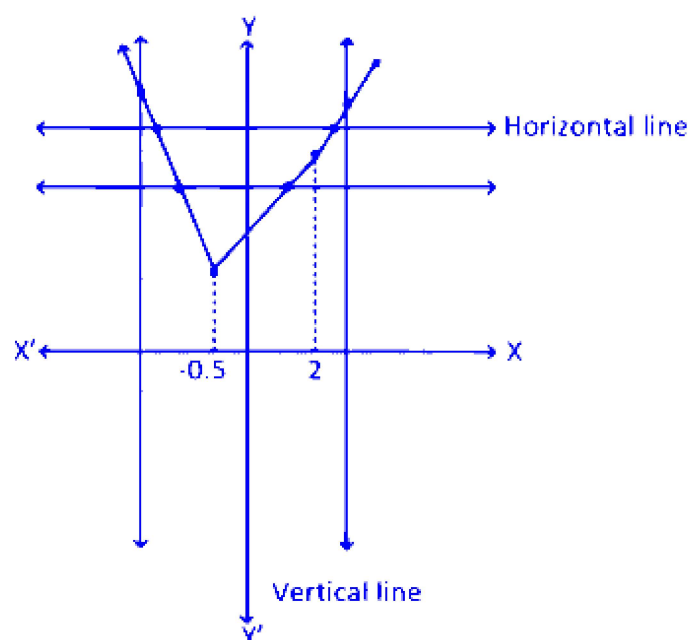


Each vertical line should intersect the graph at exactly one point. If not then function is not onto.

If a horizontal line intersects the graph more than once, the function is not one-one.

Clearly, from graph, through horizontal line two cut points implies that not one-one and through vertical line one cut point implies that onto.

Hence, function is onto but not one-one.



Question25

If $3f(x) - 2f(1/x) = x$, then $f(2) =$

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Options:

A. 1

B. $1/2$

C. 2

D. $7/2$

Answer: B

Solution:

We start with the equation:

$$3f(x) - 2f\left(\frac{1}{x}\right) = x.$$

Differentiating both sides with respect to x , we have:

$$3f'(x) - 2 \cdot f'\left(\frac{1}{x}\right) \times \left(-\frac{1}{x^2}\right) = 1.$$

This simplifies to:

$$3f'(x) + \frac{2}{x^2} f'\left(\frac{1}{x}\right) = 1.$$

First, set $x = 2$:

$$3f'(2) + \frac{2}{(2)^2} f'\left(\frac{1}{2}\right) = 1.$$

This results in:

$$3f'(2) + \frac{1}{2} f'\left(\frac{1}{2}\right) = 1.$$

Multiply throughout by 2 to clear the fraction:

$$6f'(2) + f'\left(\frac{1}{2}\right) = 2. \quad (i)$$

Next, set $x = \frac{1}{2}$:

$$3f'\left(\frac{1}{2}\right) + \frac{2}{\left(\frac{1}{2}\right)^2} f'(2) = 1.$$

This becomes:

$$3f'\left(\frac{1}{2}\right) + 8f'(2) = 1.$$

Solve for $f' \left(\frac{1}{2} \right)$:

$$f' \left(\frac{1}{2} \right) = \frac{1-8f'(2)}{3}. \quad (ii)$$

Substitute the expression for $f' \left(\frac{1}{2} \right)$ from equation (ii) into equation (i):

$$6f'(2) + \frac{1-8f'(2)}{3} = 2.$$

Multiply through by 3 to eliminate the fraction:

$$18f'(2) + 1 - 8f'(2) = 6.$$

Simplify:

$$10f'(2) = 5.$$

So, we find:

$$f'(2) = \frac{5}{10} = \frac{1}{2}.$$

Question26

The domain of the real valued function $f(x) = \log_2 \log_3 \log_5 (x^2 - 5x + 11)$ is

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Options:

A. $(2, \infty)$

B. $(-\infty, 3)$

C. $(2, 3)$

D. $(-\infty, 2) \cup (3, \infty)$

Answer: D

Solution:

Mathematics

(d) We have,

$$f(x) = \log_2 \log_3 \log_5 (x^2 - 5x + 11)$$

$\log_5$ must be positive.

$\log_5 (x^2 - 5x + 11)$ is defined if

$$x^2 - 5x + 11 > 0$$

$$\text{Now, } D = b^2 - 4ac = (-5)^2 - 4(1)(11)$$

$$= -19$$

$\therefore$ If the discriminant is negative and $a > 0$, $x^2 - 5x + 11 > 0, \forall$ real x .

Now, $\log_3$ must be positive.

$\log_3 (\log_5 (x^2 - 5x + 11))$ is defined if

$$\log_5 (x^2 - 5x + 11) > 0$$

$$\text{Since, } \log_5 (x^2 - 5x + 11) > 0$$

$$\Rightarrow x^2 - 5x + 11 > 5^0$$
$$\left[\because \log_a x = b, \text{ then } x = a^b \right]$$

$$\Rightarrow x^2 - 5x + 11 > 1$$

$$\Rightarrow x^2 - 5x + 10 > 0$$

$$D = b^2 - 4ac = (-5)^2 - 4(1)(10) = -15$$

The discriminant is negative, $x^2 - 5x + 10$ has no real roots and is always positive. $\therefore x^2 - 5x + 10 > 0, \forall$ real x

Now, $\log_2$ must be positive.

$\log_2 (\log_3 (\log_5 (x^2 - 5x + 11)))$ is defined

If

$$\log_3 (\log_5 (x^2 - 5x + 11)) > 0$$

$$\log_3 (\log_5 (x^2 - 5x + 11)) > 0$$

$$\log_5 (x^2 - 5x + 11) > 3^0$$

$$\Rightarrow \log_5 (x^2 - 5x + 11) > 1$$

$$\Rightarrow x^2 - 5x + 11 > 5$$

$$\Rightarrow x^2 - 5x + 6 > 0$$

$$(x - 2)(x - 3) > 0$$

The quadratic $x^2 - 5x + 6$ changes sign at $x = 2$ and $x = 3$.

Thus, $x^2 - 5x + 6 > 0$ for $x \in (-\infty, 2) \cup (3, \infty)$.

Thus, the domain of $f(x)$ is $(-\infty, 2) \cup (3, \infty)$.

Question 27

The range of the real valued function $f(x) = \left(\frac{x^2+2x-15}{2x^2+13x+15} \right)$ is

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Options:

A. $R - \left\{ -5, -\frac{3}{2} \right\}$

B. $R - \left\{ -5, \frac{1}{2} \right\}$

C. $R - \left\{ \frac{1}{2}, \frac{8}{7} \right\}$

D. $R - \left\{ -\frac{3}{2}, \frac{8}{7} \right\}$

Answer: C

Solution:

To find the range of the function $f(x) = \frac{x^2+2x-15}{2x^2+13x+15}$, we begin by factorizing the numerator and denominator.

Denominator Factorization:

$$2x^2 + 13x + 15 = 0$$

$$2x^2 + 10x + 3x + 15 = 0$$

$$2x(x + 5) + 3(x + 5) = 0$$

$$(x + 5)(2x + 3) = 0 \Rightarrow x = -5, -\frac{3}{2}$$

Numerator Factorization:

$$x^2 + 2x - 15 = 0$$

$$x^2 + (5x - 3x) - 15 = 0$$

$$x(x + 5) - 3(x + 5) = 0$$

$$(x + 5)(x - 3) = 0 \Rightarrow x = -5, 3$$

Domain and Range:

The domain is $(-\infty, -5) \cup (-5, -\frac{3}{2}) \cup (-\frac{3}{2}, \infty)$.

The range is $(-\infty, \frac{1}{2}) \cup (\frac{1}{2}, \frac{8}{7}) \cup (\frac{8}{7}, \infty)$, excluding $\frac{1}{2}$ and $\frac{8}{7}$, which do not correspond to any real x values for $f(x)$.

Thus, the range of the function is:

$$R - \left\{ \frac{1}{2}, \frac{8}{7} \right\}$$

Question28

$f : R \rightarrow R$ is defined by $f(x + y) = f(x) + 12y, \forall x, y \in R$. If $f(1) = 6$, then $\sum_{r=1}^n f(r) =$

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Options:

A. n^2

B. $5n^2$

C. $6n^2$

D. $\frac{3n(n+1)}{2}$

Answer: C

Solution:

Given, $f(1) = 6$

and $f(x + y) = f(x) + 12y$

Now,

$$\begin{aligned} f(2) &= f(1 + 1) = f(1) + 12(1) \\ &= 6 + 12 = 18 \end{aligned}$$

$$\begin{aligned} f(3) &= f(1 + 2) = f(1) + 12(2) \\ &= 6 + 24 = 30 \end{aligned}$$

$$\begin{aligned} f(4) &= f(1 + 3) = f(1) + 12(3) \\ &= 6 + 36 = 42 \end{aligned}$$

and, so on

$$\begin{aligned} \therefore f(n) &= 6 + (n - 1)12 \\ &= 6 + 12n - 12 = 12n - 6 \end{aligned}$$

Now, $\sum_{r=1}^n f(r) = f(1) + f(2) + f(3) + \dots + f(n)$

$$\begin{aligned} &= 6 + 18 + 30 + 42 + \dots + (12n - 6) \\ &= 6[1 + 3 + 5 + 7 + \dots + (2n - 1)] \\ &= 6n^2 \end{aligned}$$

Question29

The domain of the real valued function $f(x) = \sqrt{2+x} + \sqrt{3-x}$ is

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Options:

A. $(-2, 3)$

B. $[-2, 3)$

C. $(-2, 3]$

D. $[-2, 3]$

Answer: D

Solution:

To determine the domain of the function $f(x) = \sqrt{2+x} + \sqrt{3-x}$, we need to ensure that all expressions under the square roots are non-negative.

For $\sqrt{2+x}$ to be defined, the expression inside the square root must be non-negative:

$$2+x \geq 0 \Rightarrow x \geq -2$$

For $\sqrt{3-x}$ to be defined, the expression inside the square root must also be non-negative:

$$3-x \geq 0 \Rightarrow x \leq 3$$

Combining both conditions, the value of x must satisfy:

$$-2 \leq x \leq 3$$

Therefore, the domain of the function $f(x)$ is the closed interval $[-2, 3]$.

Question30

Let $f(x) = 3 + 2x$ and $g_n(x) = (f \circ f \circ f \circ \dots \text{in times } n)(x)$, $\forall n \in N$ if all the lines $y = g_n(x)$ pass through a fixed point (α, β) , then $\alpha + \beta =$

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Options:

A. -5

B. -4

C. -3

D. -6

Answer: D

Solution:

Let $f(x) = 3 + 2x$ and $g_n(x)$ be the composition of n functions f with itself. If all the lines $y = g_n(x)$ pass through a fixed point (α, β) , we need to find $\alpha + \beta$.

The function $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined as:

$$f(x) = 2x + 3 = g_1(x)$$

Now, find $f \circ f(x) = g_2(x)$:

$$\begin{aligned} g_2(x) &= 2(2x + 3) + 3 \\ &= 4x + 6 + 3 \\ &= 4x + 9 \end{aligned}$$

Next, find $f \circ f \circ f(x) = g_3(x)$:

$$\begin{aligned} g_3(x) &= 2(4x + 9) + 3 \\ &= 8x + 18 + 3 \\ &= 8x + 21 \end{aligned}$$

Observing a pattern, we have:

$$\begin{aligned} g_1(x) &= 2x + 3 = 2^1(x + 3) - 3 \\ g_2(x) &= 4x + 9 = 2^2(x + 3) - 3 \\ g_3(x) &= 8x + 21 = 2^3(x + 3) - 3 \\ &\vdots \\ g_n(x) &= 2^n(x + 3) - 3 \end{aligned}$$

Then, the expression becomes:

$$\begin{aligned} y &= g_n(x) = 2^n(x + 3) - 3 \\ y + 3 &= 2^n(x + 3) \end{aligned}$$

Now, consider the equation of a line passing through (α, β) :

$$y - \beta = m(x - \alpha)$$

Comparing the equations,

$$\begin{aligned} \alpha &= -3 \\ \beta &= -3 \end{aligned}$$

Thus, $\alpha + \beta = -3 + (-3) = -6$.

Question31

Let $a > 1$ and $0 < b < 1$. If $f : \mathbb{R} \rightarrow [0, 1]$ is defined by

$$f(x) = \begin{cases} a^x, & -\infty < x < 0 \\ b^x, & 0 \leq x < \infty \end{cases}, \text{ then } f(x) \text{ is}$$

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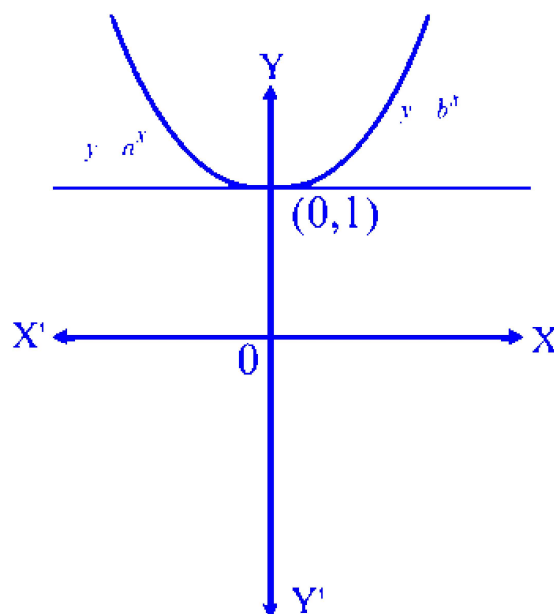
Options:

- A. a bijection
- B. one-one but not onto
- C. onto but not one-one
- D. neither one-one nor onto

Answer: D

Solution:

The graph of $f(x)$ is shown below.



∴ The function is not onto.

The function is many one as image of $f(x)$ is for every value of a and b .

Hence, the function is neither one-one nor onto.

Question32

If $P(x) = x^5 + ax^4 + bx^3 + cx^2 + dx + e$ is a polynomial such that $P(0) = 1, P(1) = 2, P(2) = 5, P(3) = 10$ and $P(4) = 17$, then $P(5) =$

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Options:

- A. 26
- B. 146
- C. 126
- D. 76

Answer: B

Solution:

We have,

$$P(x) = x^5 + ax^4 + bx^3 + cx^2 + dx + e$$

$$\because P(0) = 1$$

$$\Rightarrow 0 + 0 + 0 + 0 + 0 + e = 1$$

$$\Rightarrow e = 1$$

$$\because P(1) = 2$$

$$\Rightarrow 1 + a + b + c + d + 1 = 2$$

$$\Rightarrow a + b + c + d = 0 \dots\dots (i)$$

$$\Rightarrow 32 + 16a + 8b + 4c + 2d + 1 = 5$$

$$\Rightarrow 16a + 8b + 4c + 2d = -28$$

$$\Rightarrow 8a + 4b + 2c + d = -14 \dots\dots (ii)$$

$$\because P(3) = 10$$

$$\Rightarrow 243 + 81a + 27b + 9c + 3d + 1 = 10$$

$$\Rightarrow 81a + 27b + 9c + 3d = -234$$

$$\Rightarrow 27a + 9b + 3c + d = -78 \dots\dots (iii)$$

$$P(4) = 17$$

$$\Rightarrow 1024 + 256a + 64b + 16c + 4d + 1 = 17$$

$$\Rightarrow 256a + 64b + 16c + 4d = -1008$$

$$\Rightarrow 64a + 16b + 4c + d = -252 \dots\dots (iv)$$

From Eqs. (i), (ii), (iii) and (iv), we get

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 8 & 4 & 2 & 1 \\ 27 & 9 & 3 & 1 \\ 64 & 16 & 4 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ -14 \\ -78 \\ -252 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 8 & 4 & 2 & 1 \\ 27 & 9 & 3 & 1 \\ 64 & 16 & 4 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ -14 \\ -78 \\ -252 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 8R_1$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -4 & -6 & -7 \\ 27 & 9 & 3 & 1 \\ 64 & 16 & 4 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ -14 \\ -78 \\ -252 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 27R_1$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -4 & -6 & -7 \\ 0 & -18 & -24 & -26 \\ 64 & 16 & 4 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ -14 \\ -78 \\ -252 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - 64R_1$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -4 & -6 & -7 \\ 0 & -18 & -24 & -26 \\ 0 & -48 & -60 & -63 \end{bmatrix} \begin{bmatrix} 0 \\ -14 \\ -78 \\ -252 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - \frac{9}{2}R_2$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -4 & -6 & -7 \\ 0 & 0 & 3 & \frac{11}{2} \\ 0 & -48 & -60 & -63 \end{bmatrix} \begin{bmatrix} 0 \\ -14 \\ -15 \\ -252 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - 12R_2$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -4 & -6 & -7 \\ 0 & 0 & 3 & \frac{11}{2} \\ 0 & 0 & 12 & 21 \end{bmatrix} \begin{bmatrix} 0 \\ -14 \\ -15 \\ -84 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - 4R_3$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -4 & -6 & -7 \\ 0 & 0 & 3 & \frac{11}{2} \\ 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ -14 \\ -15 \\ -24 \end{bmatrix}$$

Thus,

$$\begin{aligned} a + b + c + d &= 0 \\ -4b - 6c - 7d &= -14 \\ 3c + \frac{11}{2}d &= -15 \\ -d &= -24 \end{aligned}$$

On solving, we get

$$a = -10, b = 35, c = -49, d = 24$$

$$\text{Now, } P(5) = 3125 + 625 \times (-10)$$

$$\begin{aligned} &+ 125 \times 35 + 25 \times (-49) + 24 \times 5 + 1 \\ &= 3125 - 6250 + 4375 - 1225 + 120 + 1 \\ &= 146 \end{aligned}$$

Question33

If a real valued function $f : [a, \infty) \rightarrow [b, \infty)$ defined by $f(x) = 2x^2 - 3x + 5$ is a bijection. Then, $3a + 2b =$

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Options:

A. 20

B. 10

C. 12

D. 6

Answer: B

Solution:

We are given a real-valued function $f : [a, \infty) \rightarrow [b, \infty)$ defined by $f(x) = 2x^2 - 3x + 5$. This function is a quadratic and is stated to be a bijection.

First, let's find the vertex of the quadratic function, as it will help determine the minimum value of the function, which is crucial for it to be a bijection.

The vertex formula for a quadratic function $ax^2 + bx + c$ is given by:

$$\left(\frac{-b}{2a}, \frac{-D}{4a}\right)$$

where $D = b^2 - 4ac$. For our function, $a = 2$, $b = -3$, and $c = 5$.

Calculating the vertex:

Calculate x -coordinate of the vertex:

$$\frac{-(-3)}{2 \times 2} = \frac{3}{4}$$

Calculate the discriminant D :

$$D = (-3)^2 - 4 \times 2 \times 5 = 9 - 40 = -31$$

Calculate y -coordinate of the vertex:

$$\frac{-(-31)}{8} = \frac{31}{8}$$

Thus, the vertex of the quadratic function is $\left(\frac{3}{4}, \frac{31}{8}\right)$.

Since $f(x) = 2x^2 - 3x + 5$ is a parabola opening upwards (as the coefficient of x^2 is positive), the minimum value of $f(x)$ occurs at the vertex. Therefore, to achieve a bijective function, the domain should start from the x -coordinate of the vertex, and the range should start from the y -coordinate of the vertex.

This implies:

$$a = \frac{3}{4}$$

$$b = \frac{31}{8}$$

To find the value of $3a + 2b$:

$$3 \times \frac{3}{4} + 2 \times \frac{31}{8} = \frac{9}{4} + \frac{62}{8} = \frac{9}{4} + \frac{31}{4} = \frac{40}{4} = 10$$

Thus, $3a + 2b = 10$.

Question34

The domain of the real valued function

$$f(x) = \frac{1}{\sqrt{\log_{0.5}(2x-3)}} + \sqrt{4-9x^2} \text{ is}$$

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Options:

A. $\left[\frac{2}{3}, \frac{3}{2}\right)$

B. Null Set

C. $\left[\frac{2}{3}, 2\right)$

D. $\left[-\frac{2}{3}, \frac{2}{3}\right]$

Answer: B

Solution:

To find the domain of the real-valued function $f(x) = \frac{1}{\sqrt{\log_{0.5}(2x-3)}} + \sqrt{4-9x^2}$, we analyze the conditions under which each part of the function is defined.

First Part: $\frac{1}{\sqrt{\log_{0.5}(2x-3)}}$

For the square root in the denominator to be defined and non-zero, the expression inside must be positive:
 $\log_{0.5}(2x-3) > 0$.

The base of the logarithm is less than 1. This means:

$$2x - 3 < 1 \Rightarrow x < 2$$

Additionally, the expression inside the logarithm must be positive:

$$2x - 3 > 0 \Rightarrow x > \frac{3}{2}$$

Second Part: $\sqrt{4-9x^2}$

The expression under the square root must be non-negative:

$$4 - 9x^2 > 0 \Rightarrow x^2 < \frac{4}{9}$$

Solving for x , we derive:

$$-\frac{2}{3} < x < \frac{2}{3}$$

Intersection of Conditions:

From the first part, we need x to satisfy both conditions: $\frac{3}{2} < x < 2$.

From the second part, x must be in the interval $-\frac{2}{3} < x < \frac{2}{3}$.

The intersection of these conditions is empty, as there is no number that simultaneously satisfies $\frac{3}{2} < x < 2$ and $-\frac{2}{3} < x < \frac{2}{3}$.

Hence, the domain of the function is a null set.

Question35

If a function $f : R \rightarrow R$ is defined by $f(x) = x^3 - x$, then f is

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Options:

- A. one-one and onto.
- B. one-one but not onto.
- C. onto but not one-one.
- D. Neither one-one nor onto.

Answer: C

Solution:

We are given the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by:

$$f(x) = x^3 - x$$

We need to determine whether this function is one-to-one (injective) and onto (surjective).

Step 1: Check if the function is one-to-one (injective)

A function is one-to-one if each element in the domain corresponds to a unique element in the codomain.

To check whether $f(x) = x^3 - x$ is one-to-one, we look at its derivative:

$$f'(x) = 3x^2 - 1$$

We find the critical points by setting $f'(x) = 0$:

$$3x^2 - 1 = 0 \Rightarrow x^2 = \frac{1}{3} \Rightarrow x = \pm \frac{1}{\sqrt{3}}$$

These critical points suggest that the function has a local maximum and minimum, which means the function is not strictly increasing or decreasing. Therefore, $f(x)$ is **not one-to-one**.

Step 2: Check if the function is onto (surjective)

A function is onto if every element of the codomain has a corresponding element in the domain. The

function $f(x) = x^3 - x$ is a cubic function. As $x \rightarrow \infty$, $f(x) \rightarrow \infty$, and as $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$.

Thus, the function covers all real values, meaning it is **onto**.

Step 3: Conclusion

Since the function is onto but not one-to-one, the correct answer is:

C: onto but not one-to-one.

Question36

If $f(x) = \sqrt{x} - 1$ and $g\{f(x)\} = x + 2\sqrt{x} + 1$ then $g(x) =$

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Options:

A.

$$(x + 2)^2$$

B.

$$(x - 2)^2$$

C.

$$(\sqrt{x} + 2)^2$$

D.

$$(\sqrt{x} - 2)^2$$

Answer: A

Solution:

To determine $g(x)$, let's first understand the given functions:

We have $f(x) = \sqrt{x-1}$. This means $f(x)$ is a function that transforms x into $\sqrt{x-1}$.

Next, the composition $g(f(x)) = x + 2x^2 + 1$ can be rewritten using substitution and simplification steps:

Begin by rewriting the expression for $g(f(x))$:

$$g(f(x)) = x + 2\sqrt{x-1} + 1$$

Recognize a perfect square identity in:

$$x + 2\sqrt{x-1} + 1 = (\sqrt{x-1} + 1)^2$$

Here, $(\sqrt{x-1} + 1)^2 = \sqrt{x-1}^2 + 2\sqrt{x-1} + 1$, which simplifies to $x - 1 + 2\sqrt{x-1} + 1 = x + 2\sqrt{x-1}$.

Notice then that:

$$g(f(x)) = (\sqrt{x-1} + 1)^2 = ((\sqrt{x-1} - 1) + 2)^2 = (f(x) + 2)^2$$

Thus, we can deduce that $g(x)$ must match the transformed output when $x = f(x)$ plus an added constant:

$$g(x) = (x + 2)^2$$

Therefore, $g(x)$ is the function that squares the input, transformed by adding 2.

Question37

For real values of x and a , if the expression $\frac{x^3 - 3x^2 - 3x + 1}{2x^2 - 3x + 1}$ assumes all real values, then

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Options:

A. $a > -1$ or $a < -1/2$

B. $-1 < a < -\frac{1}{2}$

C. $\frac{1}{2} < a < 1$

D. $a < \frac{1}{2}$ or $a > 1$

Answer: B

Solution:

Let's solve the problem such that the given expression assumes all real values.

We're looking at the expression:

$$y = \frac{x+a}{2x^2-3x+1}$$

To ensure that y can take all real values, the denominator must not be zero:

$$2x^2 - 3x + 1 \neq 0$$

This can be factored as:

$$(2x - 1)(x - 1) \neq 0$$

Thus, $x \neq 1$ and $x \neq \frac{1}{2}$.

For y to take all real values for real x , the quadratic expression $(3y + 1)^2 - 4y(y - 2) \geq 0$ must hold:

$$(3y + 1)^2 - 4[y(y - 2) - a] \geq 0$$

Simplifying the terms:

$$9y^2 + 6y + 1 - (4y^2 - 8y - 4a) \geq 0$$

This simplifies to:

$$5y^2 + 14y + 1 + 4a \geq 0$$

As a quadratic in terms of y , for it to be non-positive (invertible for any x), the discriminant must be negative:

The discriminant:

$$6^2 - 4 \times 5 \times (1 + 4a) < 0$$

$$36 - 20(1 + 4a) < 0$$

$$36 - 20 - 80a < 0$$

$$16 < 80a$$

Thus:

$$a > \frac{1}{2}$$

Therefore, the condition for a is:

$$-1 < a < -\frac{1}{2}$$

The correct understanding implies the feasible range for a .

Question38

$f(x + h) = 0$ represents the transformed equation of the equation $f(x) = x^4 + 2x^3 - 19x^2 - 8x + 60 = 0$. If this transformation removes the term containing x^3 from $f(x) = 0$, then $h =$

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Options:

A. $-1/2$

B. 1

C. 2

D. -1

Answer: A

Solution:

To find the value of h that removes the x^3 term from the polynomial $f(x) = x^4 + 2x^3 - 19x^2 - 8x + 60 = 0$ when transformed to $f(x + h) = 0$, follow these steps:

Polynomial Transformation: The transformation affects each term in the original polynomial. For the x^3 term in the polynomial, apply the transformation:

$$f(x + h) = (x + h)^4 + 2(x + h)^3 - 19(x + h)^2 - 8(x + h) + 60$$

Focus on the x^3 Terms: When expanding, consider only the terms involving x^3 :

From $(x + h)^4$:

$${}^4C_1 \cdot x^3 \cdot h = 4hx^3$$

From $2(x + h)^3$:

$$2 \times 1 \times x^3 = 2x^3$$

Collect x^3 Terms: Adding these terms together gives:

$$4hx^3 + 2x^3 = (4h + 2)x^3$$

Set the Coefficient to Zero: To eliminate the x^3 term from the polynomial, the coefficient must be zero:

$$4h + 2 = 0$$

Solve for h :

$$4h = -2 \implies h = \frac{-2}{4} = \frac{-1}{2}$$

Thus, the value of h that removes the x^3 term is $h = -\frac{1}{2}$.

Question39

$$f(x) = \log \left(\left(\frac{2x^2-3}{x} \right) + \sqrt{\frac{4x^4-11x^2+9}{|x|}} \right) \text{ is}$$

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Options:

- A. an odd function
- B. an even function
- C. a polynomial function
- D. not a function

Answer: A

Solution:

$$f(x) = \log \left(\left(\frac{2x^2-3}{x} \right) + \sqrt{\frac{4x^4-11x^2+9}{|x|}} \right)$$

$$f(-x) = \log \left(\frac{2(-x)^2-3}{(-x)} + \sqrt{\frac{4(-x)^4-11(-x)^2+9}{|-x|}} \right)$$

$$f(-x) = \log \left[\frac{2x^2-3}{-x} + \sqrt{\frac{4x^4-11x^2+9}{|x|}} \right]$$

We know that, if function $f(x)$ is an odd function. Then,

$$\begin{aligned}
& f(x) + f(-x) = 0 \\
& \log \left[\left(\frac{2x^2 - 3}{x} \right) + \sqrt{\frac{4x^4 - 11x^2 + 9}{|x|}} \right] \\
& + \log \left[\sqrt{\frac{4x^4 - 11x^2 + 9}{|x|}} - \frac{(2x^2 - 3)}{x} \right] \\
& = \log \left[\frac{4x^4 - 11x^2 + 9}{x^2} - \frac{(2x^2 - 3)^2}{x^2} \right] \\
& = \log \left[\frac{4x^4 - 11x^2 + 9 - 4x^4 - 9 + 12x^2}{x^2} \right] \\
& = \log 1 = 0
\end{aligned}$$

Question40

Let $f : R - \left\{ \frac{-1}{2} \right\} \rightarrow R$ be defined by $f(x) = \frac{x-2}{2x+1}$. If α and β satisfy the equation $f(f(x)) = -x$, then $4(\alpha^2 + \beta^2) =$

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Options:

A. 17

B. 12

C. 24

D. 34

Answer: A

Solution:

$$f(x) = \frac{x-2}{2x+1}$$

$$f(f(x)) = -x \text{ [Given]}$$

$$\begin{aligned}
&\Rightarrow \frac{f(x) - 2}{2(f(x)) + 1} = -x \\
&\Rightarrow \frac{\frac{x-2}{2x+1} - 2}{\frac{2x-4}{2x+1} + 1} = -x \\
&\Rightarrow \frac{x - 2 - 4x - 2}{2x - 4 + 2x + 1} = -x \\
&\Rightarrow \frac{-3x - 4}{4x - 3} = -x \Rightarrow \frac{3x + 4}{4x - 3} = x \\
&\Rightarrow 3x + 4 = 4x^2 - 3x \\
&\Rightarrow 4x^2 - 6x - 4 = 0 \\
&\Rightarrow 2x^2 - 3x - 2 = 0
\end{aligned}$$

Now, $\alpha + \beta = \frac{3}{2}, \alpha\beta = -1$

Now, $(\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta$

$$\Rightarrow \frac{9}{4} = \alpha^2 + \beta^2 - 2 \quad [\because \alpha\beta = -1]$$

$$\Rightarrow \alpha^2 + \beta^2 = \frac{17}{4}$$

$$\Rightarrow 4(\alpha^2 + \beta^2) = 17$$

Question41

The domain of the real valued function $f(x) = \sin \left(\log \left(\frac{\sqrt{4-x^2}}{1-x} \right) \right)$ is

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Options:

A. $(1, 4)$

B. $(-1, 1)$

C. $(-2, 1)$

D. $(-2, 4)$

Answer: C

Solution:

Given, function $f(x) = \sin \left(\log \left(\frac{\sqrt{4-x^2}}{1-x} \right) \right)$ is defined for $x \neq 1$ and $x > -2$ and also $x < 1$. Thus, the domain of above function is $D(f) = \{x \in R; x \in (-2, 1)\}$

Question42

The range of the real valued function $f(x) = \sqrt{\frac{x^2+2x+8}{x^2+2x+4}}$ is

AP EAPCET 2022 - 4th July Morning Shift

Options:

A. $\left[\sqrt{\frac{7}{3}}, \infty \right)$

B. $(0, \infty)$

C. $(1, \infty)$

D. $\left(1, \sqrt{\frac{7}{3}} \right]$

Answer: D

Solution:

$$f(x) = \sqrt{\frac{x^2+2x+8}{x^2+2x+4}}$$

$$\text{Let } y = \frac{x^2+2x+8}{x^2+2x+4} \quad [\because f(x) = \sqrt{y}]$$

$$\Rightarrow yx^2 + 2xy + 4y = x^2 + 2x + 8$$

$$\Rightarrow (y-1)x^2 + 2x(y-1) + 4y-8 = 0$$

For x to be real, discriminant of equation must be greater than equals to zero.

$$[2(y-1)]^2 - 4(y-1)(4y-8) \geq 0$$

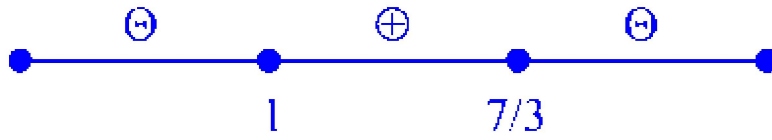
$$\Rightarrow 4(y-1)^2 - 4(y-1)4(y-2) \geq 0$$

$$\Rightarrow (y-1)^2 - (y-1)4(y-2) \geq 0$$

$$\Rightarrow (y-1)[y-1-4y+8] \geq 0$$

$$\Rightarrow (y-1)(-3y+7) \geq 0$$

$$y = 1, \frac{7}{3}$$



$$\therefore (y-1)(-3y+7) \geq 0 \Rightarrow y \in \left(1, \frac{7}{3}\right]$$

$$[\because \text{At } y=1, x^2+2x+4 \neq x^2+2x+8]$$

$$\therefore y \in \left(1, \frac{7}{3}\right]$$

$$\therefore f(x) \in \left(1, \sqrt{\frac{7}{3}}\right]$$

Question43

If $f(x) = \sqrt{2-x^2}$ and $g(x) = \log(1-x)$ are two real valued functions, then the domain of the function $(f+g)(x)$ is

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Options:

- A. $[-2, 2]$
- B. $[-2, 1)$
- C. $(-\infty, 1)$
- D. $(1, 2]$

Answer: B

Solution:

$$\begin{aligned} f(x) &= \sqrt{2-x^2} \text{ and } g(x) = \log(1-x) \\ (f+g)(x) &= f(x) + g(x) \\ &= \sqrt{2-x^2} + \log(1-x) \end{aligned}$$

$$\begin{aligned}
2 - x^2 &\geq 0 \\
\Rightarrow x^2 &\leq 2 \\
\Rightarrow |x| &\leq \sqrt{2} \\
\Rightarrow -\sqrt{2} &\leq x \leq \sqrt{2}
\end{aligned}$$

$$\text{Also, } 1 - x > 0 \Rightarrow x < 1$$

$$\text{Required domain : } -\sqrt{2} \leq x \leq \sqrt{2} \cap x < 1$$

$$\Rightarrow -\sqrt{2} \leq x < 1$$

Question44

Let $f(x) = (x + 2)^2 - 2, x \geq -2$. Then, $f^{-1}(x)$ is equal to

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Options:

A. $-\sqrt{2+x} - 2$

B. $\sqrt{2+x} + 2$

C. $\sqrt{2+x} - 2$

D. $-\sqrt{2+x} + 2$

Answer: C

Solution:

Let $f(x) = y$, then $x = f^{-1}(y)$, here

$$y = (x + 2)^2 - 2$$

or

$$y + 2 = (x + 2)^2$$

$$x + 2 = \sqrt{y + 2}$$

$$\Rightarrow x = \sqrt{y + 2} - 2$$

$$\therefore f^{-1}(y) = \sqrt{y + 2} - 2$$

$$\therefore f^{-1}(x) = \sqrt{x + 2} - 2$$

Question45

If f is the greatest integers function defined on R as $f(x) = [x]$ and g is the modulus function defined on R as $g(x) = |x|$, then the value of $(g \circ f) \left(-\frac{5}{3} \right)$ is

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Options:

A. 1

B. 2

C. 3

D. 4

Answer: B

Solution:

$$f(x) = [x], g(x) = |x|$$

$$\text{Now, } f\left(-\frac{5}{3}\right) = \left[-\frac{5}{3}\right]$$

$$= [-1.667] = -2$$

$$\text{Then, } g\left(f\left(-\frac{5}{3}\right)\right) = g(-2)$$

$$(g \circ f)\left(-\frac{5}{3}\right) = |-2| = 2$$

$$\therefore (g \circ f)\left(-\frac{5}{3}\right) = 2$$

Question46

If $f : R \rightarrow R$ and $g : R \rightarrow R$ are two functions defined by $f(x) = ax + b (a \neq 0), \forall x \in R$ and $g(x) = cx^3 + d (c \neq 0), \forall x \in R$, then $(f \circ g)^{-1}(x)$ is equal to

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Options:

A. $\left(\frac{x-ad+b}{ac}\right)^{\frac{1}{2}}$

B. $\left(\frac{x+ad-b}{ac}\right)^{\frac{1}{3}}$

C. $\left(\frac{x-ad-b}{ac}\right)^{\frac{1}{3}}$

D. $\left(\frac{x+ad+b}{ac}\right)^{\frac{1}{3}}$

Answer: C

Solution:

Given, $f(x) = ax + b$ and $g(x) = cx^3 + d$

$$\begin{aligned}f(g(x)) &= f(cx^3 + d) \\&= a(cx^3 + d) + b\end{aligned}$$

$$(f \circ g)(x) = acx^3 + ad + b \quad \dots\dots (i)$$

Let $(f \circ g)(x) = y$, then

$$x = (f \circ g)^{-1}y$$

From Eq. (i), we get

$$y = acx^3 + ad + b$$

i.e.

$$\begin{aligned}acx^3 &= y - ad - b \\x^3 &= (y - ad - b)/ac \\x &= \left[\frac{(y - ad - b)}{ac}\right]^{1/3}\end{aligned}$$

$$\therefore (f \circ g)^{-1}(y) = \left[\frac{(y - ad - b)}{ac}\right]^{1/3}$$

$$\text{or } (f \circ g)^{-1}(x) = \left[\frac{(x - ad - b)}{ac}\right]^{1/3}$$

Question47

If $f(10 - x) = 3x^2 + 4x - 5$ and $f(x) = px^2 + qx + r$, then $p + q + r$ is equal to

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Options:

A. 272

B. 274

C. 275

D. 273

Answer: B

Solution:

$$f(10 - x) = 3x^2 + 4x - 5 \text{ and } f(x) = px^2 + qx + r$$

Let $10 - x = y$, then $x = 10 - y$

$$\begin{aligned} f(y) &= 3(10 - y)^2 + 4(10 - y) - 5 \\ &= 3(100 + y^2 - 20y) + 40 - 4y - 5 \\ &= 340 + 3y^2 - 60y - 4y - 5 \\ &= 3y^2 - 64y + 335 \end{aligned}$$

$$\text{Replace } y \text{ by } x \Rightarrow f(x) = 3x^2 - 64x + 335$$

compare with $f(x) = px^2 + qx + r$, we obtain

$$\begin{aligned} p &= 3, q = -64, r = 335 \\ \Rightarrow p + q + r &= 274 \end{aligned}$$

Question48

$f(x) = \sin x + \cos x \cdot g(x) = x^2 - 1$, then $g(f(x))$ is invertible if

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Options:

A. $\frac{-\pi}{4} \leq x \leq \frac{\pi}{4}$

B. $\frac{-\pi}{2} \leq x \leq 0$

C. $\frac{-\pi}{2} \leq x \leq \pi$

D. $0 \leq x \leq \frac{\pi}{2}$

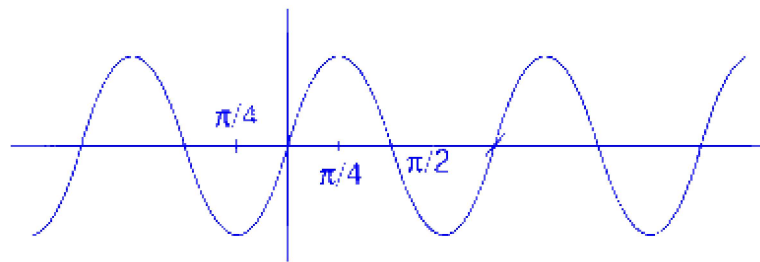
Answer: A

Solution:

$$f(x) = \sin x + \cos x$$

$$g(x) = x^2 - 1$$

$$\begin{aligned} g[f(x)] &= (\sin x + \cos x)^2 - 1 \\ &= 1 + \sin 2x - 1 = \sin 2x \end{aligned}$$



Among the given options, $\sin 2x$ is monotonous (here strictly increasing) in $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$.

Question49

If $f : z \rightarrow z$ is defined by

$$f(x) = x^9 - 11x^8 - 2x^7 + 22x^6 + x^4 - 12x^3 + 11x^2 + x - 3, \forall x \in z,$$

then $f(11)$ is equal to

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Options:

A. 7

B. 8

C. 6

D. 9

Answer: B

Solution:

$$\begin{aligned}f(x) &= x^9 - 11x^8 - 2x^7 + 22x^6 + x^4 \\&\quad - 12x^3 + 11x^2 + x - 3, \quad \forall x \in Z \\&= x^8(x - 11) - 2x^6(x - 11) + x^3(x - 11) \\&\quad - x^2(x - 11) + (x - 11) + 8 \\&= (x - 11) [x^8 - 2x^6 + x^3 - x^2 + 1] + 8\end{aligned}$$

$$\text{So, } f(11) = 0 + 8 = 8$$

Question50

Let $f(x) = x^3$ and $g(x) = 3^x$, then the quadratic equation whose roots are solutions of the equation $(f \circ g)(x) = (g \circ f)(x)$ (for $x \neq 0$) is

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Options:

A. $x^2 - 6x + 3 = 0$

B. $x^2 - 6x + 9 = 0$

C. $x^2 - x + 3 = 0$

D. $x^2 - 3 = 0$

Answer: D

Solution:

$$f(x) = x^3 \text{ and } g(x) = 3^x$$

$$\begin{aligned}\Rightarrow f[g(x)] &= g[f(x)] \\ \Rightarrow (3^x)^3 &= 3^{x^3} \Rightarrow 3^{3x} = 3^{x^3} \\ \Rightarrow 3x &= x^3 \Rightarrow x(x^2 - 3) = 0\end{aligned}$$

As, $x \neq 0$, So $x^2 - 3 = 0$

Question 51

The real valued function $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$ defined on $R/\{0\}$ is

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Options:

- A. an odd function
- B. an even function
- C. Both even and odd function
- D. Neither even nor odd function

Answer: B

Solution:

$$f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$$

$$\text{Now, } f(-x) = \frac{-x}{\frac{1}{e^x} - 1} - \frac{x}{2} + 1 = \frac{-xe^x}{1 - e^x} - \frac{x}{2} + 1$$

$$= \frac{-2xe^x - x(1 - e^x)}{2(1 - e^x)} + 1$$

$$= \frac{-2xe^x - x + xe^x}{2(1 - e^x)} + 1$$

$$= \frac{-xe^x - 2x + x}{2(1 - e^x)} + 1$$

$$= \frac{-2x + x(1 - e^x)}{2(1 - e^x)} + 1$$

$$= \frac{-x}{1 - e^x} + \frac{x}{2} + 1$$

$$= \frac{x}{e^x - 1} + \frac{x}{2} + 1 = f(x)$$

$\therefore f(x)$ is an even function.

Question52

The domain of the function $f(x) = \frac{1}{[x]-1}$, where $[x]$ is greatest integer function of x is

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Options:

A. $R - (1, 2)$

B. $R - \{1\}$

C. $R - \{0, 1\}$

D. $R - [1, 2)$

Answer: D

Solution:

$$f(x) = \frac{1}{[x]-1}$$

$f(x)$ is not defined when

$$[x] - 1 = 0$$

$$\Rightarrow [x] = 1 \Rightarrow 1 \leq x < 2$$

$$\Rightarrow x \in [1, 2)$$

$$\text{Domain } f = R - [1, 2)$$

Question53

Let $f : R \rightarrow R$ be a function defined by $f(x) = \frac{4^x}{4^x+2}$, what is the value of $f\left(\frac{1}{4}\right) + 2f\left(\frac{1}{2}\right) + f\left(\frac{3}{4}\right)$ is equal to

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Options:

A. 4

B. 3

C. 2

D. 1

Answer: C**Solution:**

$$f(x) = \frac{4^x}{4^x + 2}$$

$$f(1-x) = \frac{4^{1-x}}{4^{1-x} + 2} = \frac{\frac{4}{4^x}}{\frac{4}{4^x} + 2}$$

$$= \frac{4}{4 + 2 \cdot 4^x}$$

$$f(1-x) = \frac{2}{4^x + 2} \dots\dots (i)$$

$$\text{and } 1 - f(x) = 1 - \frac{4^x}{4^x + 2}$$

$$= \frac{4^x + 2 - 4^x}{4^x + 2} = \frac{2}{4^x + 2} = f(1-x)$$

$$\Rightarrow f(1-x) + f(x) = 1 \dots\dots (ii)$$

Put $x = \frac{1}{4}$ in Eq. (ii), we get

$$f\left(\frac{3}{4}\right) + f\left(\frac{1}{4}\right) = 1 \dots\dots (iii)$$

Now, put $x = \frac{1}{2}$ in Eq. (ii), we get

$$f\left(\frac{1}{2}\right) + f\left(\frac{1}{2}\right) = 1$$

$$\text{or } 2f\left(\frac{1}{2}\right) = 1 \dots\dots (iv)$$

Adding Eqs. (iii) and (iv), we have

$$f\left(\frac{1}{4}\right) + 2f\left(\frac{1}{2}\right) + f\left(\frac{3}{4}\right) = 2$$

Question 54

Let $f : R \rightarrow R$ and $g : R \rightarrow R$ be defined by $f(x) = 2x + 1$ and $g(x) = x^2 - 2$ determine $(g \circ f)(x)$ is equal to

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Options:

A. $2x^2 - 3$

B. $4x^2 + 4x - 1$

C. $4x^2 + 4x + 1$

D. $2x^2 - 4$

Answer: B

Solution:

Given,

$$f(x) = 2x + 1$$

$$g(x) = x^2 - 2$$

Then,

$$\begin{aligned} g \circ f(x) &= g[f(x)] \\ &= g(2x + 1) = (2x + 1)^2 - 2 \\ &= 4x^2 + 4x + 1 - 2 = 4x^2 + 4x - 1 \end{aligned}$$

Question 55

Given, the function $f(x) = \frac{a^x + a^{-x}}{2}$, ($a > 2$), then $f(x + y) + f(x - y)$ is equal to

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Options:

A. $f(x) - f(y)$

B. $f(y)$

C. $2f(x)f(y)$

D. $f(x)f(y)$

Answer: C

Solution:

Given, $f(x) = \frac{a^x + a^{-x}}{2}$

Then,

$$\begin{aligned} f(x+y) + f(x-y) &= \frac{a^{x+y} + a^{-(x+y)}}{2} \\ &+ \frac{a^{x-y} + a^{-(x-y)}}{2} \\ &= \frac{a^x \cdot a^y + a^{-x} \cdot a^{-y} + a^x \cdot a^{-y} + a^{-x} a^y}{2} \\ &= \frac{a^x (a^y + a^{-y}) + a^{-x} (a^y + a^{-y})}{2} \\ &= \frac{(a^x + a^{-x}) (a^y + a^{-y})}{2} \\ &= 2 \cdot \frac{(a^x + a^{-x})}{2} \cdot \frac{(a^y + a^{-y})}{2} = 2f(x)f(y) \end{aligned}$$

Question56

If f is a function defined on $(0, 1)$ by $f(x) = \min\{x - [x], -x - [x]\}$, then $(f \circ f \circ f \circ f)(x)$ is equal to $\rightarrow ([\cdot]$ greatest integer function)

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Options:

A. x

B. $-x$

C. $4x$

D. $2x$

Answer: A

Solution:

$$\because \{x\} = x - [x]$$

$$f : (0, 1) \rightarrow (0, 1)$$

Defined by $f(x) = \min\{x - [x], -x - [x]\}$ which is an identity function, as $x \in (0, 1)$ and $\{x\} \in (0, 1)$
Where, $\{\cdot\}$ is fractional part function.

$$\Rightarrow (f \circ f \circ f \circ f)(x) = x$$

Question 57

If $(x^2 + 5x + 5)^{x+5} = 1$, then the number of integers satisfying this equation is

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Options:

A. 2

B. 3

C. 4

D. 5

Answer: B

Solution:

$$\because (x^2 + 5x + 5)^{x+5} = 1$$

$$\text{Case I } x + 5 = 0$$

$$\Rightarrow x = -5$$

$$\text{Case II } x^2 + 5x + 5 = 1$$

$$\text{or, } x^2 + 5x + 4 = 0$$

$$\text{or, } (x + 1)(x + 4) = 0$$

$$\Rightarrow x = -1, -4$$

$x = -1, -4$ and -5 are the three integers satisfying given equation.

Question58

If $\frac{x^4}{(x-1)(x-2)} = f(x) + \frac{A}{x-1} + \frac{B}{x-2}$, then

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Options:

A. $f(x) = x^2 - 3x + 7$

B. $f(x) = x^2 + 3x + 7$

C. $A + B = 17$

D. $A - B = -18$

Answer: B

Solution:

$\frac{x^4}{(x-1)(x-2)} = \frac{x^4}{x^2-3x+2}$ is an improper fraction.

$$\begin{array}{r} x^2 - 3x + 2 \overline{) x^4 (x^2 + 3x + 7} \\ \underline{x^4 - 3x^3 + 2x^2} \\ 3x^3 - 2x^2 \\ \underline{3x^3 - 9x^2 + 6x} \\ 7x^2 - 6x \\ \underline{7x^2 - 21x + 14} \\ 15x - 14 \end{array}$$

$$\begin{aligned} \therefore \frac{x^4}{(x-1)(x-2)} &= x^2 + 3x + 7 + \frac{15x-14}{(x-1)(x-2)} \\ &= x^2 + 3x + 7 + \frac{A}{x-1} + \frac{B}{x-2} \end{aligned}$$

On comparing with given equation,

$$\frac{x^4}{(x-1)(x-2)} = f(x) + \frac{A}{x-1} + \frac{B}{x-2}, \text{ we have } f(x) = x^2 + 3x + 7$$

Question59

Which statement among the following is true?

- (i) the function $f(x) = x|x|$ is strictly increasing on $R - \{0\}$.
- (ii) the function $f(x) = \log_{(1/4)} x$ is strictly increasing on $(0, \infty)$.
- (iii) a one-one function is always an increasing function.
- (iv) $f(x) = x^{1/3}$ is strictly decreasing on R

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Options:

- A. (i)
- B. (ii)
- C. (iii)
- D. (iv)

Answer: A

Solution:

$$(i) f(x) = x|x| = \begin{cases} x(-x), & x < 0 \\ x(x), & x \geq 0 \end{cases}$$

$$f(x) = \begin{cases} -x^2, & x < 0 \\ x^2, & x \geq 0 \end{cases}$$

$$f'(x) = \begin{cases} -2x, & x < 0 \\ 2x, & x \geq 0 \end{cases}$$

$$f'(x) > 0, \forall x \in R - \{0\}$$

$\Rightarrow f(x)$ is strictly increasing on $R - \{0\}$.
